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PACKAGE STRUCTURE INCLUDING PHOTONIC PACKAGE HAVING EMBEDDED OPTICAL GLUE

Abstract

A method of forming a semiconductor package includes: bonding a first wafer to a second wafer, where the first wafer includes a plurality of electronic dies, and the second wafer includes a plurality of photonic dies; after bonding the first wafer, forming trenches in the second wafer between adjacent ones of the plurality of photonic dies; filling the trenches with an optical glue; and dicing the first wafer and the second wafer to form a plurality of photonic packages, where a photonic package of the plurality of photonic packages includes an electronic die, a photonic die bonded to the electronic die, and the optical glue, where the optical glue extends along a sidewall of the photonic package.

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Background/Summary

PRIORITY CLAIM AND CROSS-REFERENCE [0001] This application is a continuation of U.S. application Ser. No. 18/150,557, filed on Jan. 5, 2023, which claims the benefits of U.S. Provisional Application No. 63/365,136, filed on May 23, 2022 and U.S. Provisional Application No. 63/405,965, filed Sep. 13, 2022, which applications are hereby incorporated herein by reference in their entireties.

BACKGROUND

[0002] Electrical signaling and processing are one type of technique for signal transmission and processing. Optical signaling and processing have been used in increasingly more applications in recent years, particularly due to the use of optical fiber-related applications for signal transmission. [0003] Optical signaling and processing may be combined with electrical signaling and processing to provide full-fledged applications. For example, optical fibers may be used for long-range signal transmission, and electrical signals may be used for short-range signal transmission as well as processing and controlling. Accordingly, devices integrating optical components and electrical components are formed for the conversion between optical signals and electrical signals, as well as the processing of optical signals and electrical signals. Packages thus may include both optical dies (also referred to as photonic dies) including optical devices and electronic dies including electronic devices.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0005] FIGS. 1 through 6 illustrate cross-sectional views of a photonic package at various stages of manufacturing, in accordance with an embodiment.

[0006] FIGS. 7 and 8 illustrate cross-sectional views of a semiconductor package at various stages of manufacturing, in accordance with an embodiment.

[0007] FIGS. 9A, 9B, 9C, 9D, and 9E illustrate various views of a semiconductor package, in accordance with an embodiment.

[0008] FIG. 10 illustrates a cross-sectional view of a semiconductor package, in accordance with another embodiment.

[0009] FIG. 11 illustrates a cross-sectional view of a semiconductor package, in accordance with another embodiment.

[0010] FIGS. 12A, 12B, and 12C illustrate various views of a semiconductor package, in accordance with another embodiment.

[0011] FIG. 13 illustrates a cross-sectional view of a semiconductor package, in accordance with another embodiment.

[0012] FIG. 14 illustrates a cross-sectional view of a semiconductor package, in accordance with yet another embodiment.

[0013] FIG. 15 illustrates a flow chart of a method of forming a semiconductor package, in accordance with an embodiment.

DETAILED DESCRIPTION

[0014] The following disclosure provides many different embodiments, or examples, for implementing different features of the invention. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact.

[0015] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly. Throughout the description herein, unless otherwise specified, the same or similar reference numerals in different figures refer to the same or similar element formed by a same or similar formation method using a same or similar material(s).

[0016] In some embodiments, a photonic package is formed by bonding a photonic die to an electronic die, and replacing a portion of the dielectric layers of the photonic die with an optical glue. An edge coupler of the photonic die is proximate to the optical glue. An exterior sidewall of the optical glue is flush with a sidewall of the photonic package and is exposed, which allow for easy attachment of an optical fiber on the exterior sidewall of the optical glue to achieve optical coupling to the edge coupler. In some embodiments, the photonic package is embedded in a molding material with another die, and a redistribution structure is formed over the molding material. A dicing process is performed next to remove portions of the molding material to expose the exterior sidewall of the optical glue. Next, the redistribution structure is attached to a substrate, and an underfill material is formed between the redistribution structure and the substrate. The substrate has a trench formed at a topmost dielectric layer facing the redistribution structure. The trench is disposed adjacent to the optical glue, and portions of the underfill material flow into the trench, which ensures that the underfill material does not cover the exterior sidewall of the optical glue. The exposed exterior sidewall of the optical glue allows for optical coupling between a side mounted optical component (e.g., an optical fiber) and the edge coupler. The disclosed structures, by integrating both photonic dies and electronic dies in a same package, allows for co-package optics (CPO) integration, and achieves high bandwidth with ultra-low power consumption.

[0017] FIGS. 1 through 6 illustrate cross-sectional views of a photonic package 100 at various stages of manufacturing, in accordance with an embodiment.

[0018] Referring to FIG. 1, a first side (e.g., the lower side of wafer 10 in FIG. 1) of a wafer 10 is attached (e.g., bonded) to a front side of a wafer 20. The wafer 10 includes a plurality of photonic dies 120 separated by dicing regions. Locations of the dicing regions of the wafer 10 are indicated by lines 11 in FIG. 1. The wafer 20 includes a plurality of electronic dies 110 separated by dicing regions. Locations of the dicing regions of the wafer 20 are indicated by lines 21 in FIG. 1. Note that for simplicity, not all details of the photonic dies 120 and electronic dies 110 are illustrated in all of the figures. Details of the photonic die 120 and the electronic die 110 are illustrated and discussed hereinafter with reference to FIG. 6.

[0019] The wafer 10 is bonded to the wafer 20 by a suitable bonding process, such as by dielectric-to-dielectric bonding and/or metal-to-metal bonding (e.g., direct bonding, fusion bonding, oxide-to-oxide bonding, hybrid bonding, or the like). In such embodiments, covalent bonds may be formed

between oxide layers, such as the topmost dielectric layer at the first side of the wafer **10** and the topmost dielectric layer at the front side of the wafer **20**. During the bonding, metal bonding may also occur between die connectors **109** of the electronic die **110** and die connectors **127** of the photonic die **120**. In the example of FIG. **1**, dicing regions of the wafer **10** are aligned with respective dicing regions of the wafer **20** after the wafer bonding process.

[0020] Next, in FIG. **2**, conductive bumps **145** are formed on a second side (e.g., the upper side of wafer **10** in FIGS. **1** and **2**) of the wafer **10** to electrically couple to conductive features (e.g., vias, or conductive pads) of the photonic dies **120**. The conductive bumps **145**, after a subsequent dicing process to form individual photonic packages **100**, serve as the external connectors of the photonic package **100**. The conductive bumps **145** may be any suitable type of external contacts, such as ball grid arrays (BGAs), microbumps, copper pillars, a copper layer, a nickel layer, a lead free (LF) layer, an electroless nickel electroless palladium immersion gold (ENEPIG) layer, a Cu/LF layer, a Sn/Ag layer, a Sn/Pb, combinations of these, or the like.

[0021] Next, in FIG. **3**, trenches **13** (e.g., openings) are formed in the wafer **10** between adjacent photonic dies **120**. For example, the trenches **13** may be formed along the dicing regions of the wafer **10**. The trenches **13** may extend through the wafer **10** and expose the underlying wafer **20**. In the example of FIG. **3**, the bottom of the trench **13** is level with the lower surface of the wafer **10**. In some embodiments, the bottom of the trench **13** may be formed to be slightly higher or slightly lower than the lower surface of the wafer **10**.

[0022] The trenches **13** may be formed using any suitable method, such as etching (e.g., anisotropic etching), blade cutting, laser cutting, or the like. In the illustrated embodiments, each photonic die **120** has a respective trench **13** along one sidewall (e.g., the right sidewall in FIG. **3**). The trenches **13** are formed in regions of photonic dies **120** without functional circuit, such that the trenches **13** do not impact the function of the photonic dies **120**.

[0023] Next, in FIG. **4**, an optical glue **147** is formed in the trenches **13**. The optical glue **147** may fill or overfill the trenches **13**. In some embodiments, the optical glue **147** is transparent to lights within a specified range of wavelengths, which lights are used to carry optical signals processed by the photonic die **120**.

[0024] Next, in FIG. **5**, a dicing process is performed along the dicing regions of the wafers **10** and **20** to produce a plurality of individual (e.g., separate) photonic packages **100**, where each photonic package **100** includes a photonic die **120** attached (e.g., bonded) to an electronic die **110**. The dicing process may be performed using a blade, or a laser cutting tool, as examples. The dicing is performed to cut through, or cut along, the optical glue **147**, such that after the dicing process, the photonic die **120** in each photonic package **100** has the optical glue **147** embedded in the (e.g., dielectric layers of) the photonic die **120**. Due to the straight dicing path of the dicing process, an exterior sidewall **147S** of the optical glue **147** in the photonic package **100** is flush (e.g., vertically aligned) with a respective sidewall of the electronic die **110**, in some embodiments. Note that the exterior sidewall **147S** and the respective sidewall of the electronic die **110** comprise (part of) a sidewall (e.g., right sidewall in FIG. **5**) of the photonic package **100**. Details of the photonic package **100**, which includes the photonic die **120**, the electronic die **110**, and the optical glue **147**, are illustrated in FIG. **6**.

[0025] Referring now to FIG. **6**, which shows details of the photonic package **100**. Note that the photonic package **100** in FIG. **6** corresponds to the photonic package **100** in FIG. **5**, but flipped upside down. In the example of FIG. **6**, the photonic package **100** includes the electronic die **110** attached (e.g., bonded) to the photonic die **120**, with the optical glue **147** embedded in the photonic die **120** and disposed along a sidewall of the photonic package **100**.

[0026] The electronic die **110** may be, for example, semiconductor devices, dies, or chips that communicate with the photonic die **120** using electrical signals. In the illustrated embodiments, the electronic die **110** does not receive, transmit, or process optical signals. In the discussion herein, the term “electronic die” is used to distinguish from “photonic die” (e.g., **120**), which refers to a die

that can receive, transmit, or process optical signals, such as converting an optical signal into an electric signal, or vice versa. Besides optical signals, the photonic die **120** may also transmit, receive, or process electrical signals. One electronic die **110** is shown in FIG. **6**, but the photonic package **100** may include two or more electronic dies **110** in other embodiments. In some cases, multiple electronic dies **110** may be incorporated into a single photonic package **100** in order to reduce processing cost.

[0027] In some embodiments, the electronic die **110** includes a substrate **101** (e.g., a semiconductor substrate such as silicon or the like). Electronic components, such as transistors, diodes, capacitors, resistors, etc., may be formed in and/or on the substrate **101** and may be interconnected by an interconnect structures **106** formed by, for example, metallization patterns (e.g., conductive lines **105** and vias **107**) in one or more dielectric layers **103** over the substrate **101** to form an integrated circuit. The electronic die **110** further comprise pads (not illustrated), such as aluminum pads, to which external connections are made. The pads are on what may be referred to as the active side (or front side) of the electronic dies **110**. One or more passivation layers are formed at the front side of the electronic dies **110** and on portions of the pads. Die connectors **109**, such as conductive pillars (for example, comprising a metal such as copper), are formed to extend through the passivation layer(s) and are mechanically and electrically coupled to the respective pads. The die connectors **109** are electrically coupled to the integrated circuits of the electronic die **110**.

[0028] Note that in the example of FIG. **6**, portions of the dielectric layers **103** of the electronic die **110**, where no functional circuit is located, is replaced by a dielectric material **108**. The dielectric material **108** may be a gap-fill material in some embodiments, which may include silicon oxide, silicon nitride, a polymer, the like, or a combination thereof. In some embodiments, the dielectric material **108** may be a material (e.g., silicon oxide) that is substantially transparent to light at wavelengths suitable for transmitting optical signals or optical power between optical components (e.g., a grating coupler) of the photonic die **120** and a vertically-mounted optical fiber (see, e.g., **185** in FIG. **13**).

[0029] The electronic die **110** may include integrated circuits for interfacing with a photonic component **135** (e.g., a photodetector and/or a modulator) of the photonic die **120**. The electronic die **110** may include circuits for controlling the operation of the photonic components **135**. For example, the electronic die **110** may include controllers, drivers, transimpedance amplifiers, the like, or combinations thereof. The electronic die **110** may also include a central processing unit (CPU), in some embodiments. In some embodiments, the electronic die **110** includes circuits for processing electrical signals received from the photonic component **135** comprising a photodetector. The electronic die **110** may control high-frequency signaling of the photonic components **135** according to electrical signals (digital or analog) received from another device or die, in some embodiments. In some embodiments, the electronic die **110** may be an electronic integrated circuit (EIC) or the like that provides Serializer/Deserializer (SerDes) functionality. In this manner, the electronic die **110** may act as part of an I/O interface between optical signals and electrical signals within the photonic package **100**. In some embodiments, the photonic packages **100** described herein could be considered system-on-chip (SoC) or system-on-integrated-circuit (SoIC) devices.

[0030] Still referring to FIG. **6**, the photonic die **120** includes one or more dielectric layers **131**, conductive features (e.g., conductive lines **143** and vias **141**) formed in the dielectric layer **131**, and various photonic devices formed in the dielectric layers **131**, such as waveguides **133**, photonic components **135**, nitride waveguides **137** (e.g., **137A**, **137B**, and **137C**), edge coupler **139**, or the like. In addition, the photonic die **120** includes a redistribution structure **129** over the dielectric layers **131**, and conductive bumps **145** under the dielectric layers **131**.

[0031] In some embodiments, the waveguides **133** are silicon waveguides formed by patterning a silicon layer. One waveguide **133** or multiple waveguides **133** may be patterned from the silicon layer. If multiple waveguides **133** are formed, the multiple waveguides **133** may be individual

separate waveguides **133** or connected as a single continuous structure. In some embodiments, one or more of the waveguides **133** form a continuous loop.

[0032] The photonic components **135** may be integrated with the waveguides **133**, and may be formed with the waveguides **133**. The photonic components **135** may be optically coupled to the waveguides **133** to interact with optical signals within the waveguides **133**. The photonic components **135** may include, for example, photonic devices such as photodetectors and/or modulators. For example, a photodetector may be optically coupled to the waveguides **133** to detect optical signals within the waveguides **133** and generate electrical signals corresponding to the optical signals. A modulator may be optically coupled to the waveguides **133** to receive electrical signals and generate corresponding optical signals within the waveguides **133** by modulating optical power within the waveguides **133**. In this manner, the photonic components **135** facilitate the input/output (I/O) of optical signals to and from the waveguides **133**. In other embodiments, the photonic components **135** may include other active or passive components, such as laser diodes, optical signal splitters, or other types of photonic structures or devices. Optical power may be provided to the waveguides **133** by, for example, an optical fiber (see, e.g., **185** in FIG. **9A**) coupled to an external light source. Contacts **136** (e.g., copper vias) are formed to electrically couple the photonic components **135** to the redistribution structure **129** of the photonic die **120**.

[0033] Although not shown, one or more grating couplers may be integrated with the waveguides **133**, and may be formed with the waveguides **133**. The grating coupler is a photonic structure that allows optical signals and/or optical power to be transferred between the waveguide **133** and a photonic component such as a vertically-mounted optical fiber (see, e.g., optical fiber **185** in FIG. **13**) or a waveguide of another photonic system.

[0034] FIG. **6** further illustrates a plurality of nitride waveguides **137** (e.g., **137A**, **137B**, and **137C**) formed in different layers of the dielectric layers **131**. The nitride waveguides **137** may be formed by patterning silicon nitride layers. One nitride waveguide **137** or multiple nitride waveguides **137** may be formed by patterning a silicon nitride layer. If multiple nitride waveguides **137** are formed, the multiple nitride waveguides **137** may be individual separate nitride waveguides **137** or connected as a single continuous structure. In some embodiments, one or more of the nitride waveguides **137** form a continuous loop. In some embodiments, nitride waveguides **137** may include photonic structures such as grating couplers, edge couplers **139**, or couplers (e.g., mode converters) that allow optical signals to be transmitted between two nitride waveguides **137** and/or between a nitride waveguide **137** and a waveguide **133**. In the example of FIG. **6**, the nitride waveguide **137A** includes an edge coupler **139** proximate to a sidewall of the photonic die **120** facing the optical glue **147**. The edge coupler **139** allows optical signals and/or optical power to be transferred between the nitride waveguide **137A** and an external photonic component (see, e.g., optical fiber **185** in FIG. **9A**) that is horizontally mounted on a sidewall of the photonic package **100**. In the discussion herein, the waveguides **133** (e.g., silicon waveguides) and the nitride waveguides **137** (e.g., silicon nitride waveguides) may be collectively referred to as waveguides **133/137**.

[0035] The optical glue **147** embedded in the photonic die **120** has better transparency (e.g., less optical loss) for optical signals than the dielectric layers **131/121** of the photonic die **120**, thus is conducive to optical communication between the edge coupler **139** and, e.g., a horizontally mounted optical fiber (see, e.g., optical fiber **185** in FIG. **9A**). In the example of FIG. **6**, the optical glue **147** protrudes from the lower surface of the dielectric layer **131**. A surface **147T** of the optical glue **147** has a curved shape (e.g., a convex shape), which may be formed by overfilling of the trenches **13** by the optical glue **147**.

[0036] When the horizontal distances between neighboring waveguides (e.g., **133**, **137A**, **137B**, **137C**) are small, e.g., when there is lateral overlapping, and also when the vertical distances between neighboring waveguides (e.g., **133**, **137A**, **137B**, **137C**) are small, light may optically inter-couple between the neighboring waveguides (e.g., **133**, **137A**, **137B**, **137C**). Accordingly, the

light in the nitride waveguide **137A** may be optically coupled to the overlying waveguide **133** through the nitride waveguides **137B** and **137C**, thereby forming the so-called “light via” (a vertical optical coupling path similar to the vertical electrical coupling path provided by a via). Therefore, a light signal from a horizontally mounted optical fiber (e.g., mounded to an exterior sidewall of the optical glue **147** next to the edge coupler **139**) is able to be received by the edge coupler **139**, then travel vertically through the “light via” and reach the photonic component **135** (e.g., a photodetector/modulator) embedded in the waveguide **133**. The photonic component **135** performs the conversion between the optical signal and the electrical signal, which allows the electronic die **110** to process the electrical signal converted from the light signal.

[0037] FIG. **6** further illustrates the redistribution structure **129** of the photonic die **120**. The redistribution structure **129** includes one or more dielectric layers **121** and conductive features (e.g., conductive lines **123** and vias **125**) formed in the dielectric layers **121**. Die connectors **127** (e.g., copper pillars, copper pads, or the like) of the photonic die **120** are formed at the upper surface of the photonic die **120** and are electrically coupled to the conductive features of the redistribution structure **129**. In some embodiments, the dielectric layers **121** and **131** are formed of dielectric material (e.g., silicon oxide) that is substantially transparent to light at wavelengths suitable for transmitting optical signals. Note that the type of components, the number of components, the arrangement/configuration of the components illustrates in FIG. **6** is merely a non-limiting example, other types/numbers of components, and other arrangement/configuration of the components are also possible, and are fully intended to be included within the scope of the present disclosure.

[0038] FIGS. **7** and **8** illustrate cross-sectional views of a semiconductor package **150** at various stages of manufacturing, in accordance with an embodiment.

[0039] In FIG. **7**, two photonic packages **100** and an electronic die **153** are attached to a carrier **151**. The carrier **151** may be, for example, a wafer (e.g., a silicon wafer), a panel, a glass substrate, a ceramic substrate, or the like. The photonic packages **100** and the electronic die **153** may be attached to the carrier **151** using, for example, an adhesive or a release layer (not shown). The electronic die **153** may be, e.g., a CPU, an application specific integrated circuit (ASIC), a high-bandwidth memory (HBM) die, or the like. The electronic die **153** may include a substrate **154**, electronic components formed in and/or on the substrate **154**, and interconnect structures connecting the electronic components to form functional circuits of the die. Die connectors **155** of the electronic die **153** provide electrical connection to the electronic die **153**. Details are similar to those of the electronic die **110**, thus not repeated. Note that the photonic packages **100** are positioned such that for each photonic package **100**, the optical glue **147** is at an exterior sidewall (e.g., not facing the electronic die **153**) of the photonic package **100**, in order to facilitate coupling of an optical fiber to the optical glue **147** in subsequent processing.

[0040] Next, a molding material **157** is formed on the carrier **151** around the photonic packages **100** and the electronic die **153**. The molding material **157** covers the exterior sidewalls of the optical glue **147** in the photonic packages **100**. The molding material **157** may be cured by a curing process. After the molding material **157** is formed, a planarization process, such as chemical mechanical planarization (CMP), is performed to achieve a coplanar upper surface between the photonic packages **100**, the electronic die **153**, and the molding material **157**.

[0041] Next, a redistribution structure **160** is formed over the molding material **157** and electrically coupled to the photonic packages **100** and the electronic die **153**. In some embodiments, the redistribution structure **160** comprises one or more dielectric layers **161** and one or more layers of conductive features (e.g., conductive lines **163** and via **165**) formed in the dielectric layers **161**. In some embodiments, the one or more dielectric layers **161** are formed of a polymer, such as polybenzoxazole (PBO), polyimide, benzocyclobutene (BCB), or the like. In other embodiments, the dielectric layer **161** is formed of a nitride such as silicon nitride; an oxide such as silicon oxide, phosphosilicate glass (PSG), borosilicate glass (BSG), boron-doped phosphosilicate glass (BPSG),

or the like; or the like. The one or more dielectric layers **161** may be formed by any acceptable deposition process, such as spin coating, chemical vapor deposition (CVD), laminating, the like, or a combination thereof. The conductive features **163/165** may be formed of a suitable conductive material such as copper, titanium, tungsten, aluminum, or the like.

[0042] In some embodiments, the redistribution structure **160** is formed by: forming a dielectric layer **161** over the molding material **157**, forming openings in the dielectric layer **161** to expose underlying conductive features, forming a seed layer over the dielectric layer **161** and in the openings, forming a patterned photoresist with a designed pattern over the seed layer, plating (e.g., electroplating or electroless plating) the conductive material in the designed pattern and over the seed layer, and removing the photoresist and portions of seed layer on which the conductive material is not formed.

[0043] The above processing may be repeated multiple times until a target number of dielectric layers **161** and conductive features **165/163** are formed. Other ways to form the redistribution structure **160** are also possible, and are fully intended to be included within the scope of the present disclosure.

[0044] Next, conductive bumps **167** are formed over and electrically coupled to the redistribution structure **160**. The conductive bumps **167** may be any suitable type of external contacts, such as ball grid arrays (BGAs), microbumps, copper pillars, a copper layer, a nickel layer, a lead free (LF) layer, an electroless nickel electroless palladium immersion gold (ENEPIG) layer, a Cu/LF layer, a Sn/Ag layer, a Sn/Pb, combinations of these, or the like.

[0045] As skilled artisans readily appreciate, a plurality of (e.g., identical) semiconductor packages **150** may be formed simultaneously (e.g., in the same processing steps) on the carrier **151**. These semiconductor packages **150** will be separated by a subsequent dicing process to form individual, separate, semiconductor packages **150**.

[0046] Next, in FIG. **8**, the carrier **151** is removed by a carrier de-bonding process. The structure of FIG. **7** is flipped over, and the conductive bumps **167** are attached to a dicing tape (not shown). Next, a dicing process is performed along lines **148** to produce a plurality of individual (e.g., separate) semiconductor packages **150** as illustrated in FIG. **8**.

[0047] The dicing process may be performed using a blade, or a laser cutting tool, as examples. The dicing is performed to cut through, or cut along, the optical glue **147**, such that after the dicing process, the portions of molding material **157** that cover the exterior sidewalls **147S** of the optical glue **147** are removed, and the exterior sidewalls **147S** of the optical glue **147** are exposed, as illustrated in FIG. **8**.

[0048] FIGS. **9A**, **9B**, **9C**, **9D**, and **9E** illustrate various views (e.g., cross-sectional view, plan view) of a semiconductor package **200**, in accordance with an embodiment. As illustrated in FIG. **9A**, the semiconductor package **200** is formed by attaching (e.g., bonding) the semiconductor package **150** in FIG. **8** to a substrate **170**. Next, an underfill material **189** is formed to fill the gap between the substrate **170** and the semiconductor package **150**. FIG. **9B** illustrates a zoomed-in view of an area **183** in FIG. **9A**, which shows more details.

[0049] The substrate **170**, which may be a printer circuit board (PCB), includes a dielectric core **171**, which dielectric core is formed of a dielectric material such as prepreg, epoxy, silica filler, Ajinomoto build-up film (ABF), polyimide, molding compound, or the like, in some embodiments. In some embodiments, the dielectric core **171** includes bismaleimide triazine (BT) resin, FR-4 (a composite material composed of woven fiberglass cloth with an epoxy resin binder that is flame resistant), ceramic, glass, plastic, tape, film, or other supporting materials. Vias **173** are formed extending through the dielectric core **171**. In some embodiments, the vias **173** are formed by drilling through-holes in the dielectric core **171**, and forming (e.g., plating) a conductive material (e.g., copper) along sidewalls of the through-holes. After the conductive material are formed along the sidewalls of the through-holes, remaining portions of the through-holes may be filled with a dielectric material **172**, as illustrated in the example of FIG. **9A**.

[0050] Referring to FIGS. 9A and 9B, conductive features, such as conductive lines 175A, vias 175B, and conductive pads 174 (not labeled in FIG. 9A, but labeled in FIG. 9B), are formed on opposing sides of the dielectric core 171 and serve as redistribution layers to re-route electrical signals from a first location(s) of the substrate 170 to a second location(s) of the substrate 170. The conductive features are formed in a plurality of dielectric layers 177, which may be formed of a suitable dielectric material, such as ABF or prepreg. FIG. 9A further illustrates a topmost dielectric layer 179 and a bottommost dielectric layer 179 of the substrate 170. The topmost dielectric layer 179 and the bottommost dielectric layer 179 may be formed of, e.g., solder resist, as an example. Conductive pads 174 of the substrate 170 are exposed by openings in the topmost/bottommost dielectric layers 179. In some embodiments, the conductive bumps 167 of the semiconductor package 150 are aligned with respective conductive pads 174 (see FIG. 9B) of the substrate 170, and a reflow process is performed to bond the conductive bumps 167 to the respective conductive pads 174 through solder regions 176. As illustrated in FIG. 9B, the solder regions 176 completely fills these openings during the reflow process. Note that unlike the trenches 191 discussed hereinafter (where the underfill material 189 flows into), no underfill material 189 flows into these openings (since they are filled by the solder regions 176).

[0051] As illustrated in FIG. 9A, an optical fiber 185 is attached to the exterior sidewall of the optical glue 147 in the photonic package 100, and is optically coupled to the edge coupler 139 (see FIG. 9B) of the photonic package 100 to support optical communication between the photonic package 100 and an external device. FIG. 9A further illustrates a ring 181 attached to the upper surface of the substrate 170 around the semiconductor package 150 using, e.g., an adhesive (not shown). The ring 181 may be formed of a rigid material such as glass, metal, or the like, and may be used to provide structural support and to improve the planarity of the substrate 170. External connectors 178, such as solder balls, are formed at the lower surface of the substrate 170.

[0052] FIG. 9B is a zoomed-in view of an area 183 of FIG. 9A. As illustrated in FIG. 9B, the optical fiber 185 is horizontally mounted and is attached to the exterior sidewall 147S of the optical glue 147 by, e.g., an optical glue 187. The optical glue 187 may or may not be a same material as the optical glue 147. A height H1 of the optical glue 147 is between about 1 μm and about 787 μm , and a thickness W1 of the optical glue 147 is larger than 0 μm and smaller than about 5000 μm , in some embodiments.

[0053] In FIG. 9B, the surface 147T of the optical glue is a flat surface, and is level with the lower surface of the molding material 157. This flat surface 147T may be formed by the CMP process performed after forming the molding material 157 in FIG. 7. For example, if the surface 147T of the optical glue 147 in FIG. 6 protrudes further from the lower surface of the photonic die 120 than the conductive bumps 145, the CMP process performed for the molding material 157 in FIG. 7 also removes portions of the optical glue 147 to form the flat surface 147T in FIG. 9B. If, however, the surface 147T of the optical glue 147 in FIG. 6 is closer to the lower surface of the photonic die 120 than the lower surface of the conductive bumps 145 in FIG. 6, the CMP process performed for the molding material 157 in FIG. 7 may not reach the optical glue 147, and therefore, the surface 147T of the optical glue 147 may retain its shape of FIG. 6 in the final product (see FIG. 10), or may retain another shape after the molding material 157 reshapes the surface 147T (see FIG. 11).

[0054] Notably, in FIG. 9B, a trench 191 is formed in the topmost dielectric layer 179 (e.g., a solder resist layer) at a location proximate to the optical glue 147. A width W2 of the trench 191 is between about 5 μm and about 100 μm , and a depth H2 of the trench 191 is between about 10 μm and about 30 μm , in some embodiments. FIGS. 9C and 9D further illustrate the location of the trench 191 relative to the photonic package 100, the optical glue 147, and the edge coupler 139, details of which are discussed hereinafter.

[0055] As illustrated in FIG. 9B, due to the trench 191, the underfill material 189 flows into the trench 191, thus reducing a height H3 of the fillet of the underfill material 189 measured between the upper surface of the topmost dielectric layer 179 of the substrate 170 and a topmost surface of

the fillet of the underfill material **189** under (e.g., directly under) the optical glue **147**. The H3 of the fillet of the underfill material **189** may be larger than 0 μm and smaller than about 100 μm , as an example. In FIG. **9B**, the portion of the fillet of the underfill material **189** under the optical glue **147** covers (e.g., contacts and extends along) a lower sidewall of the redistribution structure **160** and exposes an upper sidewall of the redistribution structure **160**. As a result, the exterior sidewall **147S** of the optical glue **147** is completely exposed by (e.g., free of) the underfill material **189**, which ensures that the exterior sidewall **147S** of the optical glue **147** is free of obstruction for easy attachment of the optical fiber **185**. In some embodiments, the optical fiber **185** is attached to a location of the exterior sidewall **147S** such that the optical fiber **185** is pointed toward the edge coupler **139** for efficient optical coupling.

[0056] FIG. **9C** shows a top view of the semiconductor package **200**. For simplicity, not all features of the semiconductor package **200** are illustrated. FIG. **9C** illustrates the photonic packages **100**, the electronic die **153**, and the molding material **157** in the semiconductor package **150**. FIG. **9C** also illustrates the fillet of the underfill material **189** around the sidewalls of the semiconductor package **150**, the topmost dielectric layer **179** of the substrate **170**, and the ring **181** around the semiconductor package **150**. FIG. **9C** further illustrates the edge couplers **139** of the photonic package **100**, which are illustrated in phantom because they are not visible in the top view. In addition, the trenches **191** in the topmost dielectric layer **179** are also shown in phantom, because they may be covered by the underfill material **189**. To avoid cluttering, FIG. **9C** does not show the optical glue **147**. FIG. **9D**, which shows a zoomed-in view of a portion of FIG. **9C**, further illustrates the optical glue **147** in the photonic package **100** in phantom. Therefore, FIG. **9C** (or **9D**), with the edge couplers **139**, the trenches **191**, and optical glue **147** shown, may also be referred to as a plan view of the semiconductor package **200**.

[0057] As illustrated in FIG. **9C**, the trench **191** is formed next to the edge coupler **139**. In the top view of FIG. **9C**, a center axis **193** (e.g., a longitudinal axis of the edge coupler **139**/nitride waveguide **137**) of the edge coupler **139** intersects (e.g., bisects) the trench **191**. In an example embodiment, the trench **191** is positioned symmetrically around the center axis **193**. The trench **191** is spaced apart from the right sidewall of the photonic package **100** (which is also the sidewall of the semiconductor package **150**). FIG. **9C** shows two lines A-A and B-B, where the line A-A overlaps with the center axis **193** of the edge coupler **139**, and the line B-B is parallel to the line A-A, but does not intersect the trench **191**. FIG. **9B** is the cross-sectional view along line A-A, and FIG. **9E** is the cross-sectional view along line B-B.

[0058] Referring now to FIG. **9D**, which shows a zoomed-in view of a portion of FIG. **9C**. Note that in FIG. **9D**, the optical glue **147** in the photonic package **100** is shown in phantom. As illustrated in FIG. **9D**, the exterior sidewall of the optical glue **147** overlaps (is flush) with the sidewall of the photonic package **100**. In the example of FIG. **9D**, the center axis **193** of the edge coupler **139** intersects (e.g., bisects) both the optical glue **147** and the trench **191**. In some embodiments, a vertical offset D1 between the center axis **193** and the low sidewall of the trench **191** in FIG. **9D** is between about 100 μm and about 5 mm. A width D3 of the fillet of the underfill material **189** may be between about 20 μm and about 5 mm. A horizontal offset D2 between a center axis **195** (e.g., a longitudinal axis of the trench **191** in FIG. **9D**) of the trench **191** and a corresponding sidewall of the photonic package **100** facing the trench **191** is larger than 0 μm and smaller than about 2 mm, in some embodiments.

[0059] FIG. **9E** is the cross-sectional view of the semiconductor package **200** along line B-B in FIG. **9C**. The trench **191** in the topmost dielectric layer **179** of the substrate **170** is not in the cross-sectional view of FIG. **9E**. Note that since there is no trench **191** formed in the portion of the topmost dielectric layer **179** shown in FIG. **9E**, the fillet of the underfill material **189** shown in FIG. **9E** (which is disposed outside the trench **191**) extends higher along the sidewall of the redistribution structure **160** than the fillet of the underfill material **189** shown in FIG. **9B** (which extends inside the trench **191**). For example, the fillet of the underfill material **189** in FIG. **9E** may

completely cover the sidewall of the redistribution structure **160**, as shown in FIG. **9E**, or may even cover the lower sidewall of the photonic die **120**, as shown by the dashed line **189'** in FIG. **9E**. In other words, without the trench **191**, the fillet of the underfill material **189** in FIG. **9B** may cover at least portion of the exterior sidewall of the optical glue **147**, thus hindering the attachment of the optical fiber **185**.

[0060] FIG. **10** illustrates a cross-sectional view of a portion of a semiconductor package **200A**, in accordance with another embodiment. The semiconductor package **200A** is similar to the semiconductor package **200** in FIG. **9B**, but with a different shape for the surface **147T** of the optical glue **147**. In FIG. **10**, the surface **147T** of the optical glue **147** is recessed from the lower surface of the molding material **157**, and is a curved (e.g., convex) surface. As discussed above, the shape of the surface **147T** may retain its original shape in FIG. **6**, if the CMP process for the molding material **157** does not reach the optical glue **147**.

[0061] FIG. **11** illustrates a cross-sectional view of a portion of a semiconductor package **200B**, in accordance with another embodiment. The semiconductor package **200B** is similar to the semiconductor package **200** in FIG. **9B**, but with a different shape for the surface **147T** of the optical glue **147**. In FIG. **11**, the surface **147T** of the optical glue **147** is recessed from the lower surface of the molding material **157**, and is a curved (e.g., concave) surface. The concave shape for the surface **147T** may be caused by formation of the molding material **157** and the thermal energy transferred to the optical glue **147** during formation of the molding material **157**, which shapes or reshapes the surface **147T** of the optical glue **147**. The CMP process for the molding material **157** does not reach the optical glue **147**, and therefore, does not produce the flat surface **147T** as illustrated in FIG. **9B**.

[0062] FIGS. **12A**, **12B**, and **12C** illustrate various views of a portion of a semiconductor package **200C**, in accordance with another embodiment. The semiconductor package **200C** is similar to the semiconductor package **200** in FIG. **9B**, but with the trench **191** extending under (e.g., directly under) the photonic package **100**. Unlike FIG. **9B**, where the trench **191** is disposed outside of the lateral extents of the photonic package **100**, the trench **191** in FIG. **12A** has a portion that is disposed within the lateral extents of the photonic package **100**. In some embodiments, a first sidewall of the trench **191** is disposed within boundaries (e.g., defined by sidewalls) of the photonic package **100**, and a second opposing sidewall of the trench **191** is disposed outside the boundaries of the photonic package **100**. A lateral offset **D4** between the first sidewall of the trench **191** and a respective sidewall (e.g., a closest sidewall) of the photonic package **100** is larger than 0 mm and smaller than 0.5 mm, in some embodiments.

[0063] FIG. **12B** shows a top view of the semiconductor package **200C**. The top view of the semiconductor package **200C** is similar to that in FIG. **9C**, but the trenches **191** extend into boundaries of the photonic package **100**. FIG. **12C** shows a zoomed-in view of a portion of FIG. **12B**. The optical glue **147** is also shown in FIG. **12C** in phantom. As illustrated in FIG. **12C**, the trench **191** extends into boundaries of the photonic package **100**, and may also overlap with at least portions of the edge coupler **139**. FIG. **12C** also illustrates the lateral offset **D4** shown in FIG. **12A**.

[0064] FIG. **13** illustrates a cross-sectional view of a semiconductor package **200D**, in accordance with another embodiment. The semiconductor package **200D** is similar to the semiconductor package **200** of FIG. **9A**, but with one of the photonic packages **100** in FIG. **9A** replaced by a photonic package **100A**, and with the ring **181** in FIG. **9A** replaced by a lid **201**. The lid **201** has openings in its top portion to allow attachment of, e.g., optical fibers **185** to the photonic packages **100/100A**. For example, a fiber array unit (FAU) **203** is attached to the photonic package **100** for optical coupling between a vertically mounted optical fiber **185** and the edge coupler **139** of the photonic package **100**. Furthermore, another vertically mounted optical fiber **185** is attached to the photonic package **100A** and is optically coupled to a photonic component (e.g., a grating coupler) of the photonic package **100A**. The lid **201** may be attached to the substrate **170** by, e.g., an adhesive material. A center portion of the lid **201** may contact the photonic packages **100/100A** and

the electronic die **153**, either directly or through a thermal interface material (TIM), in order to facilitate heat dissipation.

[0065] In some embodiments, the photonic package **100A** similar to the photonic package **100** of FIG. **6**, but with the nitride waveguides **137**, the edge coupler **139**, and the dielectric layers **131** in which the nitride waveguides **137** are formed replaced by a redistribution structure similar to the redistribution structure **129** of FIG. **6**. In addition, a grating coupler may be integrated in the waveguides **133** of the photonic die of the photonic package **100A**. FIG. **13** further illustrates a micro lens **209** formed in the substrate of the electronic die of the photonic package **100A**. An optical fiber **185** is vertically mounted to the backside of the photonic package **100A** over the micro lens **209** using an optical glue **187**, and is optically coupled to the grating coupler in the waveguide **133** of the photonic die of the photonic package **100A**.

[0066] The FAU **203** is attached to the backside of the photonic package **100** using an adhesive material **207**, and to the exterior sidewall of the optical glue **147** using an optical glue **205**. A vertically mounted optical fiber **185** is attached to an upper surface of the FAU **203** using an optical glue **187**. The FAU **203** provides an interface to enable optical coupling between the edge coupler **139** of the photonic die of the photonic package **100** and the vertically mounted optical fiber **185** attached to the FAU **203**.

[0067] FIG. **14** illustrates a cross-sectional view of a semiconductor package **200E**, in accordance with yet another embodiment. The semiconductor package **200E** is similar to the semiconductor package **200D** of FIG. **13**, but with the photonic package **100** on the right also replaced by the photonic package **100A**.

[0068] Embodiments may achieve advantages. For example, by embedding the optical glue **147** in the photonic die **120**, the efficiency of optical coupling between the edge coupler **139** and the external optical fiber **185** is improved. In addition, the trench **191** in the substrate **170** ensures that the underfill material **189** does not extend high along the sidewall of the photonic package **100**, and ensures that the exterior sidewall of the optical glue **147** is exposed for easy attachment of, or coupling to, optical fiber **185**. The disclosed structure and method allow for optical coupling with both edge couplers **139** (e.g., through the optical glue **147** at the sidewall of the photonic package) and grating couplers (e.g., through a vertically mounted optical fiber) of the photonic packages, as illustrated in FIGS. **13** and **14**. The disclosed semiconductor packages (e.g., **150**, **200**), by integrating both photonic dies and electronic dies in a same package, allows for co-package optics (CPO) integration, and achieves high bandwidth with ultra-low power consumption.

[0069] FIG. **15** illustrates a flow chart of a method **1000** of forming a semiconductor package, in accordance with some embodiments. It should be understood that the embodiment method shown in FIG. **15** is merely an example of many possible embodiment methods. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. For example, various steps as illustrated in FIG. **15** may be added, removed, replaced, rearranged, or repeated.

[0070] Referring to FIG. **15**, at block **1010**, a first wafer is bonded to a second wafer, wherein the first wafer comprises a plurality of electronic dies, and the second wafer comprises a plurality of photonic dies. At block **1020**, after bonding the first wafer, trenches are formed in the second wafer between adjacent ones of the plurality of photonic dies. At block **1030**, the trenches are filled with an optical glue. At block **1040**, the first wafer and the second wafer are diced to form a plurality of photonic packages, wherein a photonic package of the plurality of photonic packages comprises an electronic die, a photonic die bonded to the electronic die, and the optical glue, wherein the optical glue extends along a sidewall of the photonic package.

[0071] In accordance with an embodiment, a method of forming a semiconductor package includes: bonding a first wafer to a second wafer, wherein the first wafer comprises a plurality of electronic dies, and the second wafer comprises a plurality of photonic dies; after bonding the first wafer, forming trenches in the second wafer between adjacent ones of the plurality of photonic dies; filling the trenches with an optical glue; and dicing the first wafer and the second wafer to form a

plurality of photonic packages, wherein a photonic package of the plurality of photonic packages comprises an electronic die, a photonic die bonded to the electronic die, and the optical glue, wherein the optical glue extends along a sidewall of the photonic package. In an embodiment, the first wafer has first dicing regions, and the second wafer has second dicing regions, wherein after bonding the first wafer to the second wafer, the first dicing regions are aligned with respective ones of the second dicing regions. In an embodiment, the trenches are formed along the second dicing regions. In an embodiment, the trenches are formed to extend through the second wafer. In an embodiment, an exterior sidewall of the optical glue of the photonic package is flush with the sidewall of the photonic package. In an embodiment, the photonic die comprises a first waveguide and an edge coupler optically coupled to the first waveguide, wherein the edge coupler is proximate to a first sidewall of the photonic die facing the optical glue. In an embodiment, the optical glue is formed to contact and extend along the first sidewall of the photonic die. In an embodiment, the method further includes: attaching the photonic package to a carrier; attaching a die to the carrier laterally adjacent to the photonic package; forming a molding material over the carrier around the photonic package and the die; and forming a first redistribution structure over and electrically coupled to the die and the photonic package. In an embodiment, the method further comprises, after forming the first redistribution structure, performing a dicing process, wherein the dicing process removes portions of the molding material and exposes the exterior sidewall of the optical glue. In an embodiment, the method further comprises: after performing the dicing process, attaching the first redistribution structure to a first side of a substrate; and forming an underfill material between the first redistribution structure and the substrate, wherein the underfill material exposes the exterior sidewall of the optical glue. In an embodiment, the method further comprises attaching an optical fiber to the exterior sidewall of the optical glue. In an embodiment, the substrate comprises: a dielectric core; a second redistribution structure at a first side of the dielectric core; and a solder resist layer over the second redistribution structure at the first side of the substrate, wherein there is a trench in the solder resist layer, wherein attaching the first redistribution structure comprises aligning the optical glue with the trench in the solder resist layer such that in a plan view, a center axis of the edge coupler intersects the optical glue and the trench in the solder resist layer.

[0072] In accordance with an embodiment, a method of forming a semiconductor package includes: attaching an photonic package and a semiconductor die to a carrier, wherein the photonic package comprises an electronic die, a photonic die bonded to the electronic die, and an optical glue embedded in the photonic die, wherein a first sidewall of the optical glue is flush with a sidewall of the electronic die; forming a molding material over the carrier around the photonic package and the semiconductor die, the molding material covering the first sidewall of the optical glue; forming a redistribution structure over the molding material; and after forming the redistribution structure, removing a portion of the molding material to expose the first sidewall of the optical glue. In an embodiment, the method further comprises, after removing the portion of the molding material: attaching the redistribution structure to a substrate; and forming an underfill material between the redistribution structure and the substrate. In an embodiment, the substrate comprises a dielectric core, a conductive feature extending along a first surface of the dielectric core, and a dielectric layer on the conductive feature, wherein there is a trench in the dielectric layer, wherein after attaching the redistribution structure, the optical glue is adjacent to the trench. In an embodiment, the photonic die comprises an edge coupler for optical communication, wherein the edge coupler is adjacent to a sidewall of the photonic die facing the optical glue, wherein after attaching the redistribution structure, the edge coupler, the optical glue, and the trench are aligned along a same line in a plan view. In an embodiment, the method further comprises, after forming the underfill material, optically coupling an external optical fiber to the edge coupler through the optical glue.

[0073] In accordance with an embodiment, a semiconductor package includes: a redistribution structure; a photonic package attached to a first side of the redistribution structure, the photonic package comprising an electronic die, a photonic die bonded to the electronic die, and an optical

glue embedded in the photonic die, wherein a first sidewall of the optical glue is flush with a sidewall of the electronic die; a molding material on the first side of the redistribution structure, wherein the molding material partially surrounds the photonic package and exposes the first sidewall of the optical glue; a substrate attached to a second opposing side of the redistribution structure; and an underfill material between the substrate and the redistribution structure, wherein the underfill material exposes the first sidewall of the optical glue. In an embodiment, the substrate comprises a dielectric core, a conductive feature extending along a first surface of the dielectric core, and a dielectric layer on the conductive feature, wherein there is a trench in the dielectric layer. In an embodiment, the photonic die comprises an edge coupler for optical communication, wherein the edge coupler is adjacent to a sidewall of the photonic die facing the optical glue, wherein in a plan view, a center axis of the edge coupler intersects the optical glue and the trench. [0074] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A method of forming a semiconductor package, the method comprising: bonding a first wafer to a second wafer, wherein the first wafer comprises a first electronic die and a second electric die, and the second wafer comprises a first photonic die and a second photonic die; after bonding the first wafer, forming a trench in the second wafer between the first photonic die and the second photonic die; filling the trench with an optical glue; and performing a first dicing process through the first wafer and the second wafer to form a photonic package comprising the first electronic die, the first photonic die, and the optical glue, wherein after the first dicing process, an exterior sidewall of the optical glue is flush with a sidewall of the photonic package.
2. The method of claim 1, wherein the first wafer has a first dicing region, and the second wafer has a second dicing region, and wherein bonding the first wafer to the second wafer comprises aligning the first dicing region with the second dicing region.
3. The method of claim 1, wherein an exterior sidewall of the optical glue of the photonic package is flush with the sidewall of the photonic package.
4. The method of claim 1, wherein the first photonic die comprises a first waveguide and an edge coupler optically coupled to the first waveguide, wherein the edge coupler is proximate to a first sidewall of the photonic die facing the optical glue.
5. The method of claim 1 further comprising: bonding the photonic package to a redistribution structure; encapsulating the photonic package in an encapsulant; and performing a second dicing process through the encapsulant, wherein the second dicing process exposes an exterior sidewall of the optical glue.
6. A photonic package comprising: an electronic die; a photonic die bonded to the electronic die by dielectric-to-dielectric and metal-to-metal bonding, wherein the photonic die comprises an optical waveguide and an edge coupler optically connected to the edge coupler; and an optical glue on a sidewall of the photonic die, wherein an exterior sidewall of the optical glue is aligned with an exterior sidewall of the electronic die, and wherein the edge coupler is disposed at the sidewall of the photonic die.
7. The photonic package of claim 6, wherein the electronic die comprises: a semiconductor substrate; an interconnect structure on the semiconductor substrate; and a dielectric material along

- a sidewall of the interconnect structure, wherein the dielectric material extends continuously from the semiconductor substrate to the photonic die.
- 8.** The photonic package of claim 7, wherein the dielectric material further extends continuously from the semiconductor substrate to the optical glue.
- 9.** The photonic package of claim 6, wherein the photonic die further comprises external connectors on a surface of the photonic die that is opposite to the electronic die.
- 10.** A semiconductor package comprising: a redistribution structure; a photonic package attached to a first side of the redistribution structure, the photonic package comprising an electronic die, a photonic die bonded to the electronic die, and an optical glue extending along a first sidewall of the photonic die, wherein a first sidewall of the optical glue is flush with a first sidewall of the electronic die; and a molding material extending along a second sidewall of the photonic die and a second sidewall of the electronic die, the first sidewall of the photonic die being opposite to the second sidewall of the photonic die, and the first sidewall of the electronic die being opposite to the second sidewall of the electronic die.
- 11.** The semiconductor package of claim 10 further comprising: a substrate attached to a second side of the redistribution structure, the second side of the redistribution structure being opposite to the first side of the redistribution structure; and an underfill material between the substrate and the redistribution structure, wherein the underfill material exposes a sidewall of the optical glue.
- 12.** The semiconductor package of claim 11, wherein the substrate comprises a dielectric core, a conductive feature extending along a first surface of the dielectric core, and a dielectric layer on the conductive feature, wherein the underfill extends into a trench in the dielectric layer.
- 13.** The semiconductor package of claim 12, wherein the trench surrounds the photonic package in a plan view.
- 14.** The semiconductor package of claim 12, wherein the photonic die comprises an edge coupler at the first sidewall of the photonic die, wherein in a plan view, a line along a center axis of the edge coupler intersects the optical glue and the trench.
- 15.** The semiconductor package of claim 10 further comprising an optical fiber directly attached to the optical glue.
- 16.** The semiconductor package of claim 10, wherein the optical glue is level with a lateral surface of the molding material.
- 17.** The semiconductor package of claim 10, wherein an interface between a lateral surface of the optical glue and the molding material is convex.
- 18.** The semiconductor package of claim 10, wherein an interface between a lateral surface of the optical glue and the molding material is concave.
- 19.** The semiconductor package of claim 10, wherein the electronic die comprises a dielectric material extending continuously from the optical glue to a semiconductor substrate of the electronic die.
- 20.** The semiconductor package of claim 10, wherein a lateral surface of the optical glue protrudes from a lateral surface of the photonic die.
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